Lesson Planning

Complete Lesson Plan Book

Class: Class 11 – FSc Pre-Medical

Subject: Biology

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

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Unit 1: The Cell

Learning Objectives

• Define and explain cell structure in the context of The Cell.

• Differentiate between cell membrane and cytoplasm with suitable examples.

• Solve conceptual problems related to nucleus accurately.

• Apply the ideas of The Cell to everyday situations involving cell organelles.

Key Concepts

• Meaning, definition and scope of cell structure.

• Important properties, types and examples of cell membrane.

• Relationship between cytoplasm and nucleus.

• Practical applications of The Cell, especially cell organelles.

• Key terms and a summary of the main points of The Cell.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce The Cell with a daily-life example.
2. Explain cell structure on the board with definitions and clear diagrams.
3. Discuss cell membrane and cytoplasm using simple, relatable examples.
4. Involve the students through short questions on nucleus.
5. Solve an exercise question on cell organelles together with the class.
6. Summarize the lesson and highlight the key points of The Cell.

Classroom Activities

• Group discussion: students compare cell membrane with cytoplasm.

• Board activity: solve a question on nucleus in pairs.

• Quick quiz on the key terms of The Cell.

Assessment Questions

1. Define cell structure.
2. Differentiate between cell membrane and cytoplasm.
3. Explain with an example: nucleus.
4. Write a short note on cell organelles.

Homework

• Read The Cell from the book and solve the exercise questions.

• Prepare short notes on cell structure and cell membrane.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Biological Molecules

Learning Objectives

• Define and explain carbohydrates in the context of Biological Molecules.

• Differentiate between proteins and lipids with suitable examples.

• Solve conceptual problems related to nucleic acids accurately.

• Apply the ideas of Biological Molecules to everyday situations involving water.

Key Concepts

• Meaning, definition and scope of carbohydrates.

• Important properties, types and examples of proteins.

• Relationship between lipids and nucleic acids.

• Practical applications of Biological Molecules, especially water.

• Key terms and a summary of the main points of Biological Molecules.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Biological Molecules with a daily-life example.
2. Explain carbohydrates on the board with definitions and clear diagrams.
3. Discuss proteins and lipids using simple, relatable examples.
4. Involve the students through short questions on nucleic acids.
5. Solve an exercise question on water together with the class.
6. Summarize the lesson and highlight the key points of Biological Molecules.

Classroom Activities

• Group discussion: students compare proteins with lipids.

• Board activity: solve a question on nucleic acids in pairs.

• Quick quiz on the key terms of Biological Molecules.

Assessment Questions

1. Define carbohydrates.
2. Differentiate between proteins and lipids.
3. Explain with an example: nucleic acids.
4. Write a short note on water.

Homework

• Read Biological Molecules from the book and solve the exercise questions.

• Prepare short notes on carbohydrates and proteins.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Enzymes

Learning Objectives

• Define and explain enzyme structure in the context of Enzymes.

• Differentiate between enzyme action and factors affecting enzymes with suitable examples.

• Solve conceptual problems related to enzyme inhibition accurately.

• Apply the ideas of Enzymes to everyday situations involving industrial uses.

Key Concepts

• Meaning, definition and scope of enzyme structure.

• Important properties, types and examples of enzyme action.

• Relationship between factors affecting enzymes and enzyme inhibition.

• Practical applications of Enzymes, especially industrial uses.

• Key terms and a summary of the main points of Enzymes.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Enzymes with a daily-life example.
2. Explain enzyme structure on the board with definitions and clear diagrams.
3. Discuss enzyme action and factors affecting enzymes using simple, relatable examples.
4. Involve the students through short questions on enzyme inhibition.
5. Solve an exercise question on industrial uses together with the class.
6. Summarize the lesson and highlight the key points of Enzymes.

Classroom Activities

• Group discussion: students compare enzyme action with factors affecting enzymes.

• Board activity: solve a question on enzyme inhibition in pairs.

• Quick quiz on the key terms of Enzymes.

Assessment Questions

1. Define enzyme structure.
2. Differentiate between enzyme action and factors affecting enzymes.
3. Explain with an example: enzyme inhibition.
4. Write a short note on industrial uses.

Homework

• Read Enzymes from the book and solve the exercise questions.

• Prepare short notes on enzyme structure and enzyme action.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Bioenergetics

Learning Objectives

• Define and explain photosynthesis in the context of Bioenergetics.
• Differentiate between respiration and ATP with suitable examples.
• Solve conceptual problems related to chloroplast accurately.
• Apply the ideas of Bioenergetics to everyday situations involving mitochondria.

Key Concepts

• Meaning, definition and scope of photosynthesis.
• Important properties, types and examples of respiration.
• Relationship between ATP and chloroplast.
• Practical applications of Bioenergetics, especially mitochondria.
• Key terms and a summary of the main points of Bioenergetics.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Bioenergetics with a daily-life example.
2. Explain photosynthesis on the board with definitions and clear diagrams.
3. Discuss respiration and ATP using simple, relatable examples.
4. Involve the students through short questions on chloroplast.
5. Solve an exercise question on mitochondria together with the class.
6. Summarize the lesson and highlight the key points of Bioenergetics.

Classroom Activities

• Group discussion: students compare respiration with ATP.
• Board activity: solve a question on chloroplast in pairs.
• Quick quiz on the key terms of Bioenergetics.

Assessment Questions

1. Define photosynthesis.
2. Differentiate between respiration and ATP.
3. Explain with an example: chloroplast.
4. Write a short note on mitochondria.

Homework

• Read Bioenergetics from the book and solve the exercise questions.
• Prepare short notes on photosynthesis and respiration.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Nutrition

Learning Objectives

• Define and explain autotrophic nutrition in the context of Nutrition.

• Differentiate between heterotrophic nutrition and human digestive system with suitable examples.

• Solve conceptual problems related to vitamins accurately.

• Apply the ideas of Nutrition to everyday situations involving malnutrition.

Key Concepts

• Meaning, definition and scope of autotrophic nutrition.

• Important properties, types and examples of heterotrophic nutrition.

• Relationship between human digestive system and vitamins.

• Practical applications of Nutrition, especially malnutrition.

• Key terms and a summary of the main points of Nutrition.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Nutrition with a daily-life example.
2. Explain autotrophic nutrition on the board with definitions and clear diagrams.
3. Discuss heterotrophic nutrition and human digestive system using simple, relatable examples.
4. Involve the students through short questions on vitamins.
5. Solve an exercise question on malnutrition together with the class.
6. Summarize the lesson and highlight the key points of Nutrition.

Classroom Activities

• Group discussion: students compare heterotrophic nutrition with human digestive system.

• Board activity: solve a question on vitamins in pairs.

• Quick quiz on the key terms of Nutrition.

Assessment Questions

1. Define autotrophic nutrition.
2. Differentiate between heterotrophic nutrition and human digestive system.
3. Explain with an example: vitamins.
4. Write a short note on malnutrition.

Homework

• Read Nutrition from the book and solve the exercise questions.

• Prepare short notes on autotrophic nutrition and heterotrophic nutrition.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Gaseous Exchange

Learning Objectives

• Define and explain stomata in the context of Gaseous Exchange.

• Differentiate between human respiratory system and breathing mechanism with suitable examples.

• Solve conceptual problems related to oxygen transport accurately.

• Apply the ideas of Gaseous Exchange to everyday situations involving respiratory disorders.

Key Concepts

• Meaning, definition and scope of stomata.

• Important properties, types and examples of human respiratory system.

• Relationship between breathing mechanism and oxygen transport.

• Practical applications of Gaseous Exchange, especially respiratory disorders.

• Key terms and a summary of the main points of Gaseous Exchange.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Gaseous Exchange with a daily-life example.
2. Explain stomata on the board with definitions and clear diagrams.
3. Discuss human respiratory system and breathing mechanism using simple, relatable examples.
4. Involve the students through short questions on oxygen transport.
5. Solve an exercise question on respiratory disorders together with the class.
6. Summarize the lesson and highlight the key points of Gaseous Exchange.

Classroom Activities

• Group discussion: students compare human respiratory system with breathing mechanism.

• Board activity: solve a question on oxygen transport in pairs.

• Quick quiz on the key terms of Gaseous Exchange.

Assessment Questions

1. Define stomata.
2. Differentiate between human respiratory system and breathing mechanism.
3. Explain with an example: oxygen transport.
4. Write a short note on respiratory disorders.

Homework

• Read Gaseous Exchange from the book and solve the exercise questions.

• Prepare short notes on stomata and human respiratory system.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Transport

Learning Objectives

• Define and explain human heart in the context of Transport.
• Differentiate between blood vessels and blood composition with suitable examples.
• Solve conceptual problems related to lymphatic system accurately.
• Apply the ideas of Transport to everyday situations involving transport in plants.

Key Concepts

• Meaning, definition and scope of human heart.
• Important properties, types and examples of blood vessels.
• Relationship between blood composition and lymphatic system.
• Practical applications of Transport, especially transport in plants.
• Key terms and a summary of the main points of Transport.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Transport with a daily-life example.
2. Explain human heart on the board with definitions and clear diagrams.
3. Discuss blood vessels and blood composition using simple, relatable examples.
4. Involve the students through short questions on lymphatic system.
5. Solve an exercise question on transport in plants together with the class.
6. Summarize the lesson and highlight the key points of Transport.

Classroom Activities

• Group discussion: students compare blood vessels with blood composition.
• Board activity: solve a question on lymphatic system in pairs.
• Quick quiz on the key terms of Transport.

Assessment Questions

1. Define human heart.
2. Differentiate between blood vessels and blood composition.
3. Explain with an example: lymphatic system.
4. Write a short note on transport in plants.

Homework

• Read Transport from the book and solve the exercise questions.
• Prepare short notes on human heart and blood vessels.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Homeostasis

Learning Objectives

• Define and explain osmoregulation in the context of Homeostasis.
• Differentiate between thermoregulation and human kidney with suitable examples.
• Solve conceptual problems related to nephron accurately.
• Apply the ideas of Homeostasis to everyday situations involving excretion.

Key Concepts

• Meaning, definition and scope of osmoregulation.
• Important properties, types and examples of thermoregulation.
• Relationship between human kidney and nephron.
• Practical applications of Homeostasis, especially excretion.
• Key terms and a summary of the main points of Homeostasis.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Homeostasis with a daily-life example.
2. Explain osmoregulation on the board with definitions and clear diagrams.
3. Discuss thermoregulation and human kidney using simple, relatable examples.
4. Involve the students through short questions on nephron.
5. Solve an exercise question on excretion together with the class.
6. Summarize the lesson and highlight the key points of Homeostasis.

Classroom Activities

• Group discussion: students compare thermoregulation with human kidney.
• Board activity: solve a question on nephron in pairs.
• Quick quiz on the key terms of Homeostasis.

Assessment Questions

1. Define osmoregulation.
2. Differentiate between thermoregulation and human kidney.
3. Explain with an example: nephron.
4. Write a short note on excretion.

Homework

• Read Homeostasis from the book and solve the exercise questions.
• Prepare short notes on osmoregulation and thermoregulation.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Coordination and Control

Learning Objectives

• Define and explain human nervous system in the context of Coordination and Control.
• Differentiate between neuron and reflex action with suitable examples.
• Solve conceptual problems related to hormones accurately.
• Apply the ideas of Coordination and Control to everyday situations involving plant hormones.

Key Concepts

• Meaning, definition and scope of human nervous system.
• Important properties, types and examples of neuron.
• Relationship between reflex action and hormones.
• Practical applications of Coordination and Control, especially plant hormones.
• Key terms and a summary of the main points of Coordination and Control.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Coordination and Control with a daily-life example.
2. Explain human nervous system on the board with definitions and clear diagrams.
3. Discuss neuron and reflex action using simple, relatable examples.
4. Involve the students through short questions on hormones.
5. Solve an exercise question on plant hormones together with the class.
6. Summarize the lesson and highlight the key points of Coordination and Control.

Classroom Activities

• Group discussion: students compare neuron with reflex action.
• Board activity: solve a question on hormones in pairs.
• Quick quiz on the key terms of Coordination and Control.

Assessment Questions

1. Define human nervous system.
2. Differentiate between neuron and reflex action.
3. Explain with an example: hormones.
4. Write a short note on plant hormones.

Homework

• Read Coordination and Control from the book and solve the exercise questions.
• Prepare short notes on human nervous system and neuron.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: Reproduction

Learning Objectives

• Define and explain asexual reproduction in the context of Reproduction.

• Differentiate between sexual reproduction and human reproductive system with suitable examples.

• Solve conceptual problems related to plant reproduction accurately.

• Apply the ideas of Reproduction to everyday situations involving seed germination.

Key Concepts

• Meaning, definition and scope of asexual reproduction.

• Important properties, types and examples of sexual reproduction.

• Relationship between human reproductive system and plant reproduction.

• Practical applications of Reproduction, especially seed germination.

• Key terms and a summary of the main points of Reproduction.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Reproduction with a daily-life example.
2. Explain asexual reproduction on the board with definitions and clear diagrams.
3. Discuss sexual reproduction and human reproductive system using simple, relatable examples.
4. Involve the students through short questions on plant reproduction.
5. Solve an exercise question on seed germination together with the class.
6. Summarize the lesson and highlight the key points of Reproduction.

Classroom Activities

• Group discussion: students compare sexual reproduction with human reproductive system.

• Board activity: solve a question on plant reproduction in pairs.

• Quick quiz on the key terms of Reproduction.

Assessment Questions

1. Define asexual reproduction.
2. Differentiate between sexual reproduction and human reproductive system.
3. Explain with an example: plant reproduction.
4. Write a short note on seed germination.

Homework

• Read Reproduction from the book and solve the exercise questions.

• Prepare short notes on asexual reproduction and sexual reproduction.

Time Allocation

4–6 periods of 40 minutes each